# Debolina Tapadar

Computer Science Undergraduate

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## Professional Summary

Computer Science undergraduate at DIT University with interests in Artificial Intelligence, Machine Learning, Computer Vision, Web Development, Blockchain, and Data Analytics. Experienced in developing AI- and web-based applications, completing internships, participating in hackathons, and leading editorial initiatives within the ACM Student Chapter. Passionate about building impactful technology solutions while continuously expanding technical expertise through hands-on projects and continuous learning.

## Education

**DIT University** 2024 – 2028
Bachelor of Technology (B.Tech), Computer Science Engineering

**Delhi Public School**
Higher Secondary Education

**Nirmala Convent School**
Secondary Education

## Experience

**Machine Learning Intern** SmartED Innovations

- Completed a Machine Learning internship focused on Artificial Intelligence fundamentals and practical model development.
- Applied Python-based machine learning techniques to assignments involving data preprocessing, model implementation, and evaluation.
- Strengthened understanding of supervised learning workflows and analytical problem-solving.

**Strategic Outreach Intern** Younity.in

- Contributed to strategic outreach initiatives supporting youth engagement and entrepreneurship under the Startup India initiative.
- Collaborated with teams to improve outreach activities and community engagement.
- Received a Certificate of Recognition for contributions during the internship.

**Editorial Head** DIT University ACM Student Chapter

- Lead editorial planning, technical content creation, and event documentation.
- Coordinate newsletters, promotional material, and chapter communications.

**Member** DIT University ACM Student Chapter

- Participate in workshops, technical events, and community initiatives.

## Projects

**Medical Image Segmentation Framework**

- Developed medical image segmentation models using U-Net, U-Net++, and SwinUNet for image analysis tasks.
- Applied Ant Colony Optimization (ACO), Differential Evolution (DE), and Grey Wolf Optimization (GWO) to optimize model performance.
- Evaluated models using Dice Score, IoU, Precision, Recall, F1-Score, and Accuracy.

**PIHU – Personal Editorial Blog**

- Designed, developed, and deployed a responsive editorial-style blogging website using HTML, CSS, JavaScript, and GitHub Pages.
- Published analytical articles covering cinema, fashion, psychology, culture, and creative writing.
- Focused on clean UI design, structured content, and personal branding.

**MEDMI – Health Tracker**

- Designed a healthcare management platform using Firebase as the backend.
- Planned secure patient information management and health tracking features.
- Focused on scalable architecture and user-friendly interface design.

**Global Economy Youth Survey Website**

- Developed and deployed a responsive survey website using HTML, CSS, JavaScript, and GitHub Pages.
- Designed an intuitive user interface to collect insights on youth contributions to the global economy.
- Strengthened frontend development and responsive web design skills.

**Additional Projects**

- Blockchain-Based Voting System
- Netflix Recommendation System
- Campus Club Event Management Platform

## Technical Skills

**Programming Languages:** Python, Java, C, JavaScript, MATLAB

**Web Technologies:** HTML, CSS, JavaScript, Firebase

**Artificial Intelligence:** Machine Learning, Deep Learning, Computer Vision, Image Segmentation

**Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, NumPy, Pandas

**Optimization Techniques:** Ant Colony Optimization (ACO), Differential Evolution (DE), Grey Wolf Optimization (GWO)

**Tools:** Git, GitHub, VS Code, MATLAB, LaTeX, Overleaf

## Certifications

- Machine Learning – SmartED Innovations
- Artificial Intelligence & Machine Learning – Pregrad
- Netflix Recommendation Model Project – Pregrad

- Deloitte Australia – Data Analytics Job Simulation (Forage)
- Goldman Sachs – Excel Skills for Business Job Simulation (Forage)
- Accenture North America – Data Analytics and Visualization Job Simulation (Forage)
- Tata Group – Data Visualisation: Empowering Business with Effective Insights (Forage)
- Stock Marketing and Crypto – Acmegrade
- Human Resources – Acmegrade

## Leadership & Activities

- Editorial Head, DIT University ACM Student Chapter
- AWS Community Day Dehradun 2025 Participant
- Hackathon 4.0, UPES – Team Participant

## Languages

English Hindi Bengali